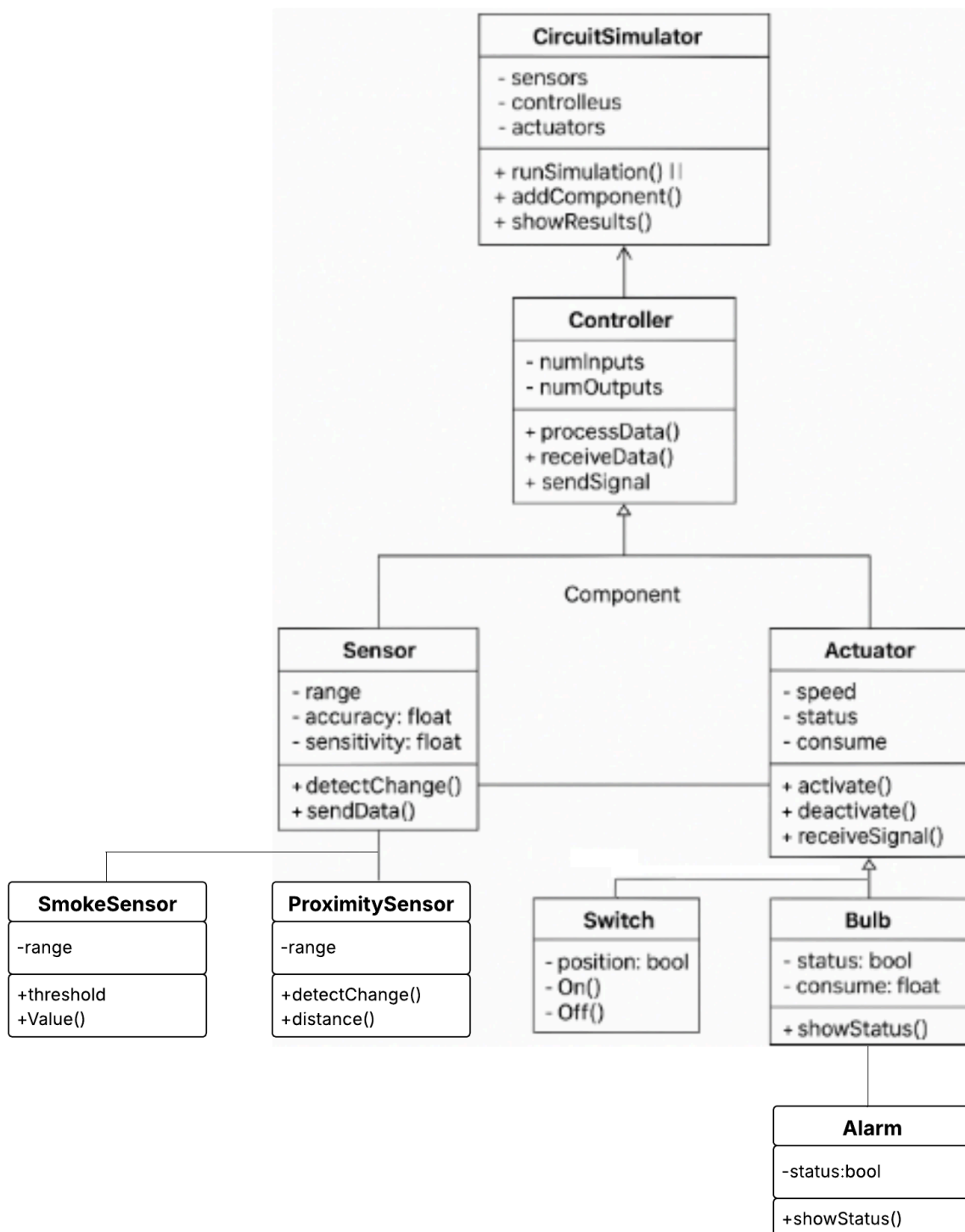


UML



Usage Instructions

Running the Application

Requirements

Python 3.10+

PyQt5 installed

Run command

`python main.py`

Creating a Circuit

Open the simulator window.

Use the “Add Component” button to create:

Switches

Bulbs

Sensors

Controllers

Components appear visually in the GUI.

Connecting Components

Select “Connect Components”.

Choose source and destination components.

Valid connections are enforced by the business logic:

Sensors → Controller

Controller → Actuator

Invalid connections show an error popup.

Running the Simulation

Press Run Simulation.

If the controller receives an activation signal from a sensor:

Connected actuators change status (e.g., bulbs turn ON).

Results are visually shown on the GUI.

Saving the Circuit

Click “Save Circuit”.

Choose a filename.

A JSON file is generated containing all components and connections.

Loading a Circuit

Click “Load Circuit”.

Select a .json file previously saved.

The GUI reconstructs all components automatically.